

Topic 10.1:

Model Assessment, Models Comparison and Model Deployment

Honest Assessment

By Dr Lee How Chinh

Objectives

- Describe the purpose of honest assessment.

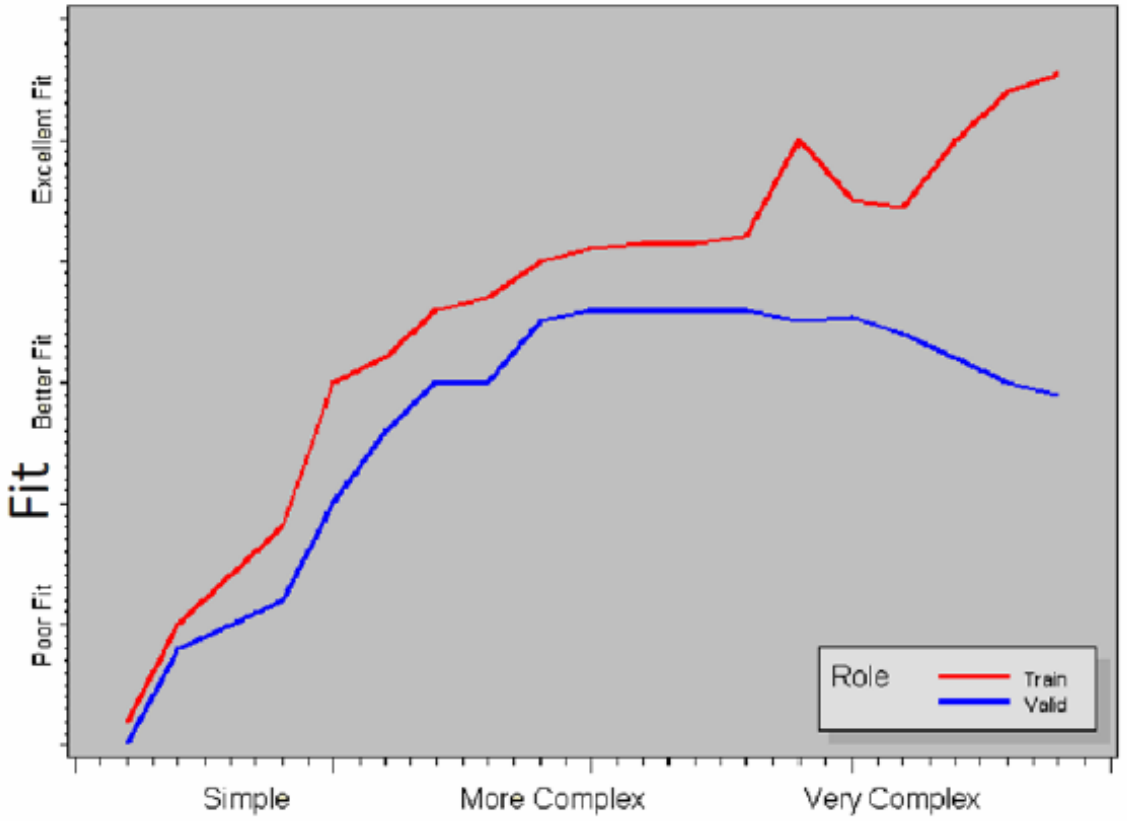
Honest Assessment

Fit versus Complexity

- The purpose of predictive modelling is generalization, which is the performance of predictions on new data.
- Evaluating the model on the same data that the model was fit on usually leads to an optimistically biased assessment.
- Data splitting is used to correct the optimism bias.
- To compare many models, an appropriate fit statistic (or statistics) must be selected and measured on the validation data set.

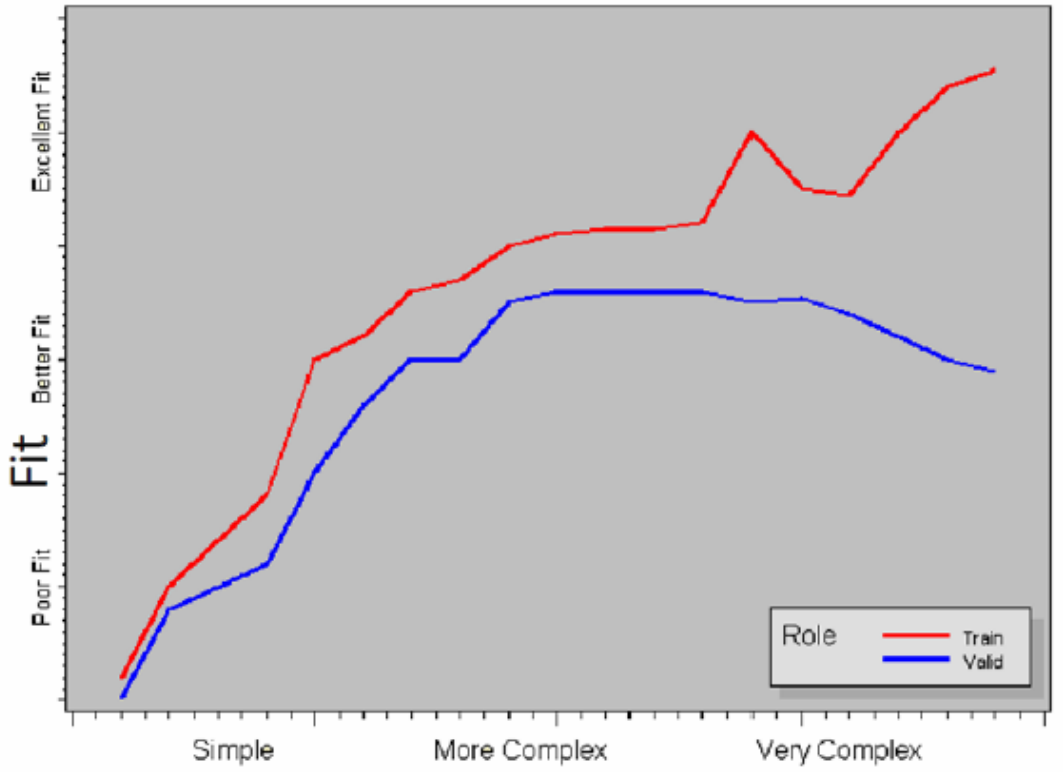
Fit versus Complexity

- Normally, model performance follows a straightforward trend.
- The fit on the training data improves as the complexity increases.
- After a certain degree of complexity, the fit improves due to the model identifying noises in the training data set.



Fit versus Complexity

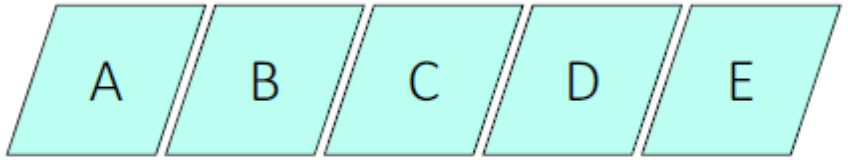
- The overfitting patterns in training data are not likely to be repeated in the validation data or in future observations.
- Therefore, the models of varying complexity fits on the validation data,
- and the model with the complexity that has the highest validation fit statistic is selected.



K-Fold Cross Validation

K-Fold Cross Validation

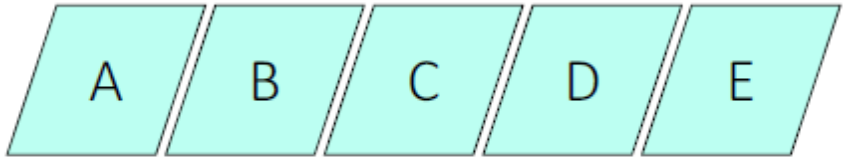
- When the data set is small, data splitting can be a costly technique.
- The performance measures on the small validation data set can be unreliable because of high variability.
- For small and moderate data sets, K-fold cross validation is a better strategy.



	Train	Validate
1)	BCDE	A
2)	ACDE	B
3)	ABDE	C
4)	ABCE	D
5)	ABCD	E

K-Fold Cross Validation

- For example, in five-fold cross validation, the data is split into five equal sets.
- The entire modelling process is repeated on each four-fifths of the data and uses the remaining one-fifth for assessment.
- The five assessments are then averaged.



	Train	Validate
1)	BCDE	A
2)	ACDE	B
3)	ABDE	C
4)	ABCE	D
5)	ABCD	E

Lesson Summary

- One of the most important steps in predictive modelling is to assess the performance of the model.
- To correct the optimism bias, the models with different complexity are evaluated on the validation data set.
- The best performance model on the validation data set is selected.
- The K-fold cross validation is used when the data set is too small to split into training and validation data sets.

How do we measure the performances of models?



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Topic 10.2

Model Assessment and Models Comparison

Confusion Matrix & Measurement Statistics

By Dr Lee How Chinh



Objectives

- Construct a Confusion Matrix.
- Explain the model performance measures derived from the Confusion Matrix.

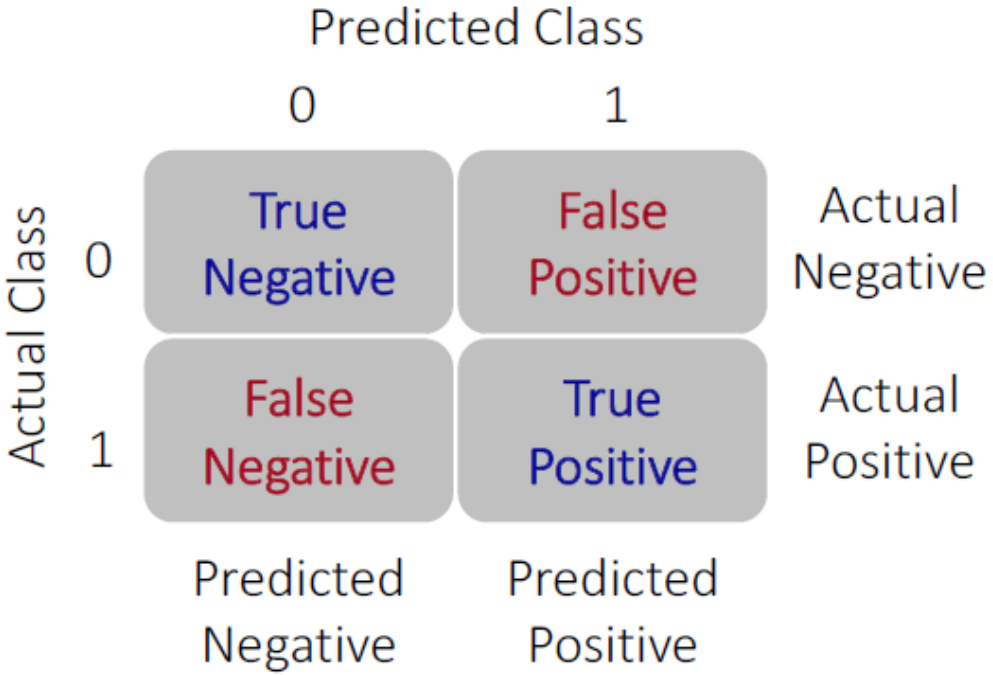
Confusion Matrix

Confusion Matrix

- Supervised classification does not usually conclude with estimated posterior probability.
- An allocation rule corresponds to a threshold value of the posterior probability is used to classified the binary cases.
- The fundamental assessment tool for the binary target is the confusion matrix.

Confusion Matrix

- The confusion matrix is a crosstabulation of the actual and predicted classes.
- It quantifies the confusion of the classifier.
- The event of interest is often called a positive.



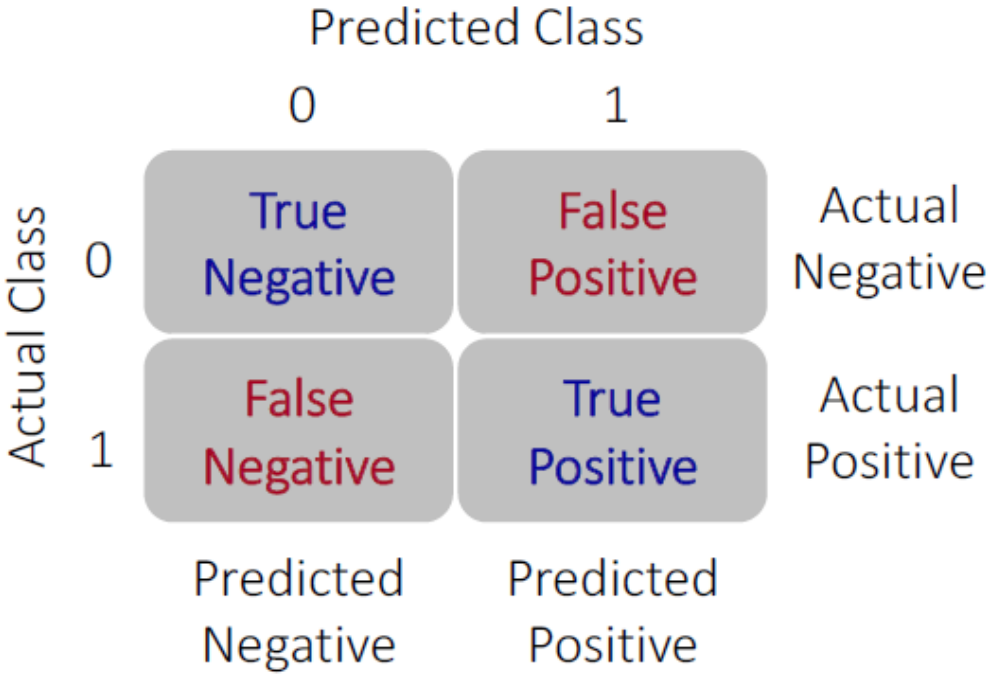
Confusion Matrix

- The simplest performance statistics are **accuracy**,

$$\text{accuracy} = \frac{\text{true positive and negatives}}{\text{total cases}}$$

- and **error rate or misclassification rate**,

$$\text{error rate} = \frac{\text{false positive and negatives}}{\text{total cases}}$$



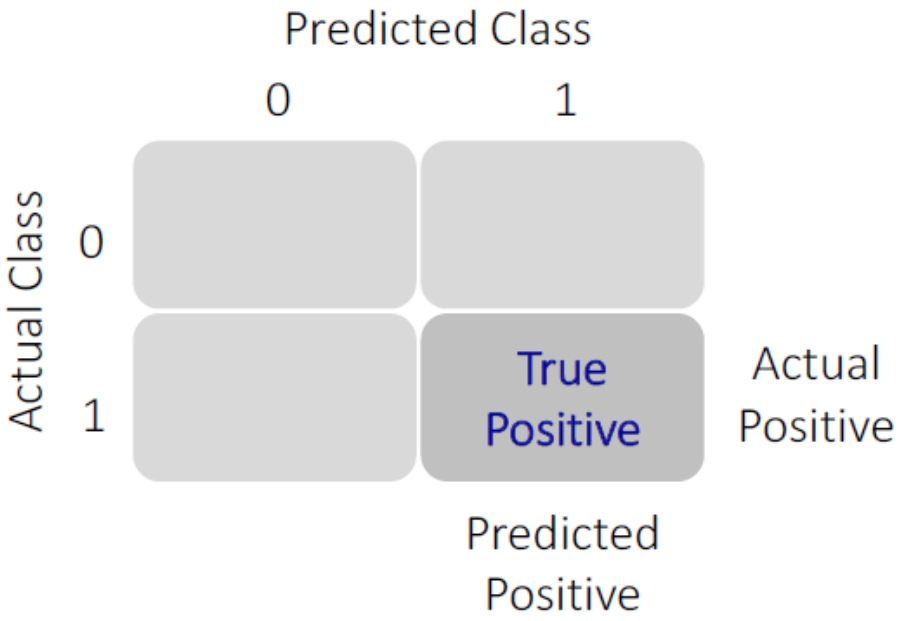
Confusion Matrix

- Two specialized measures of classifier performance are **sensitivity**,

$$\text{sensitivity} = \frac{\text{true positives}}{\text{total actual positives}}$$

- and **positive predicted value** (PV+),

$$\text{positive predicted value} = \frac{\text{true positives}}{\text{total predicted positives}}$$



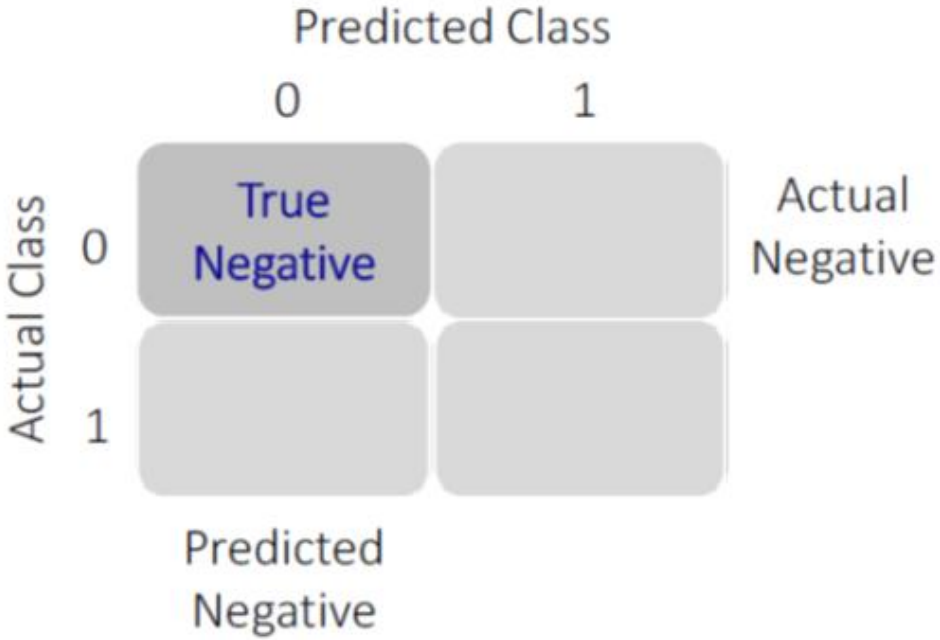
Confusion Matrix

- The analogs to these measures for true negatives are **specificity**,

$$\text{specificity} = \frac{\text{true negatives}}{\text{total actual negatives}}$$

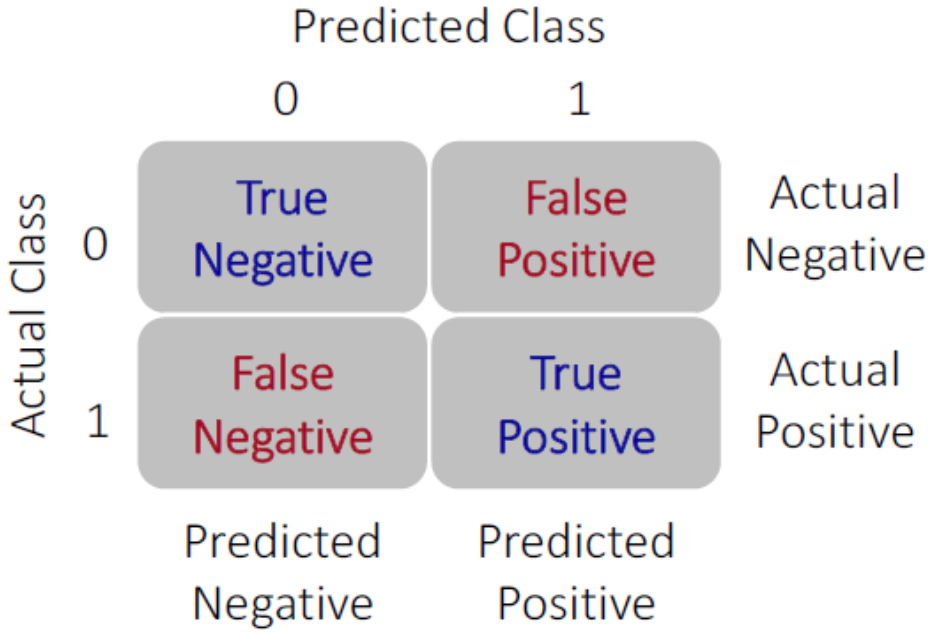
- and **negative predicted value** (PV-),

$$\text{negative predicted value} = \frac{\text{true negatives}}{\text{total predicted negatives}}$$



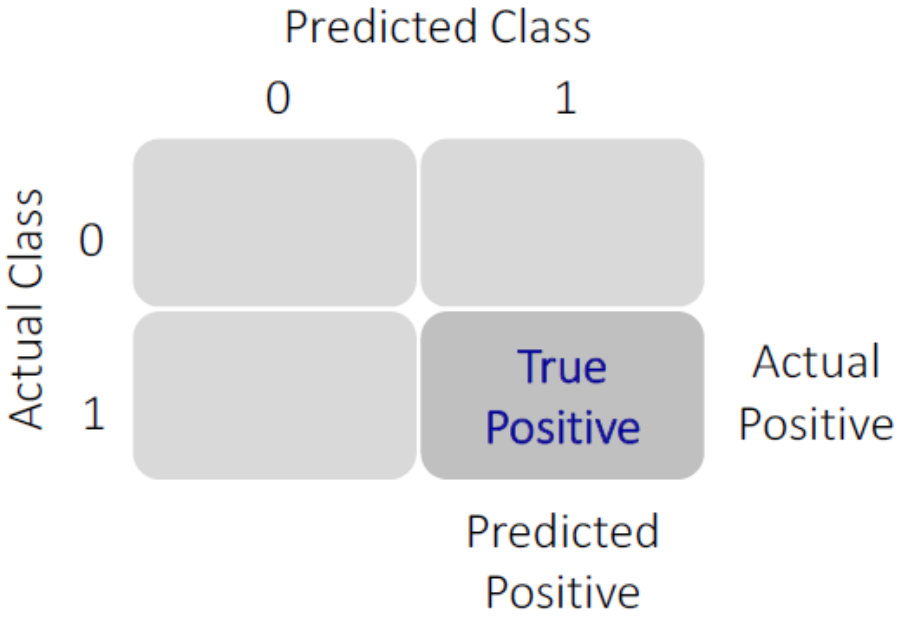
Confusion Matrix

- Ideally, you want large values for all these statistics.
- The context of the problem determines which of these measures is the primary concern.



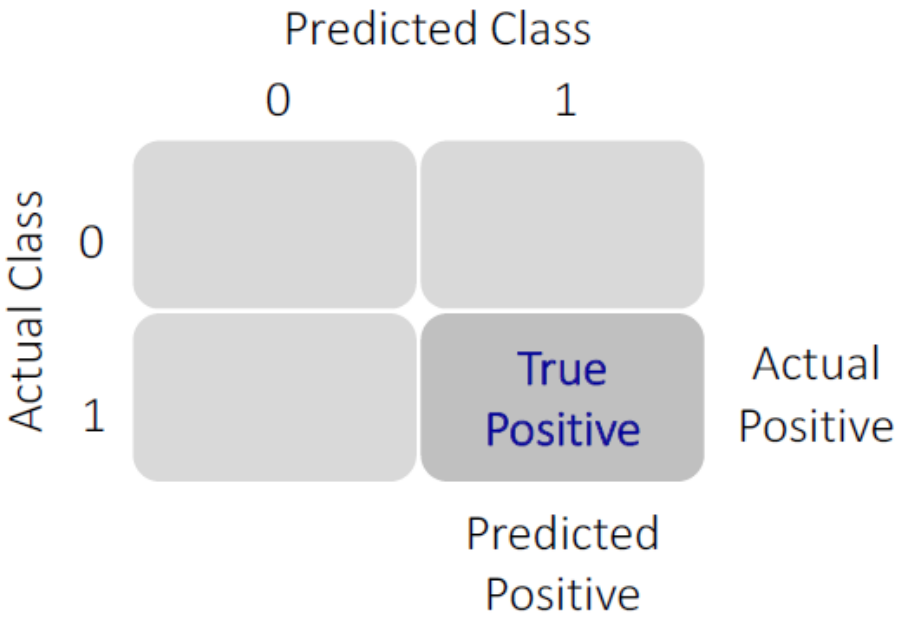
Confusion Matrix

- For example,
- a database marketer might be most concerned with PV+ because it gives the response rate for the customers that receive an offer.



Confusion Matrix

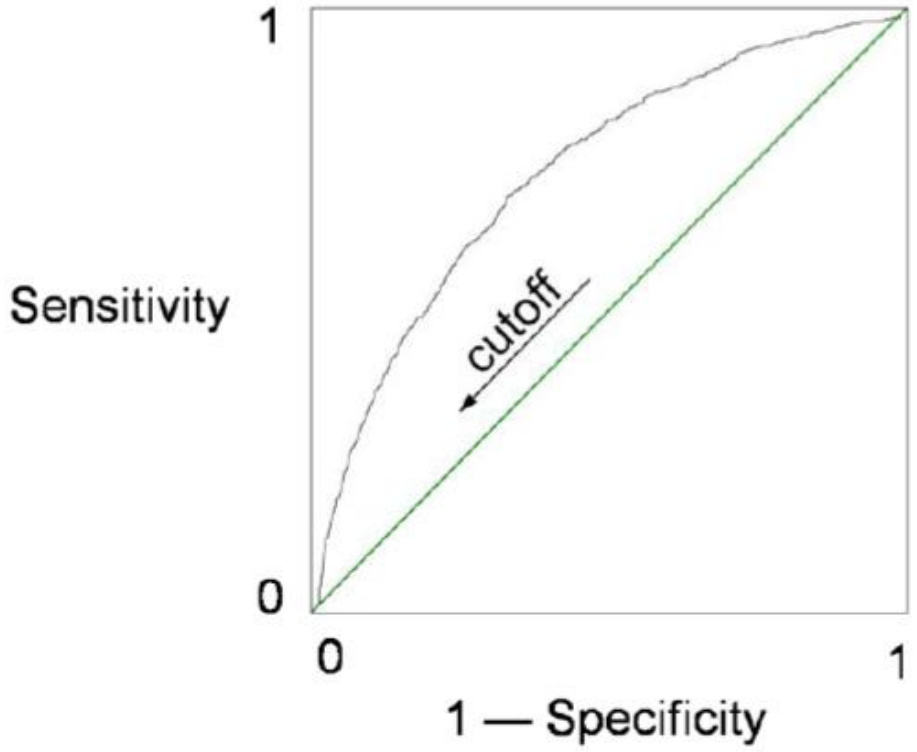
- The sensitivity is the proportion of true positives over the total actual positives. A fraud investigator might be most concerned with sensitivity, specifically about how many actual positives have been correctly predicted as the consequences of missing out on the actual positives can be very serious in fraud detection.



Measurement Statistics

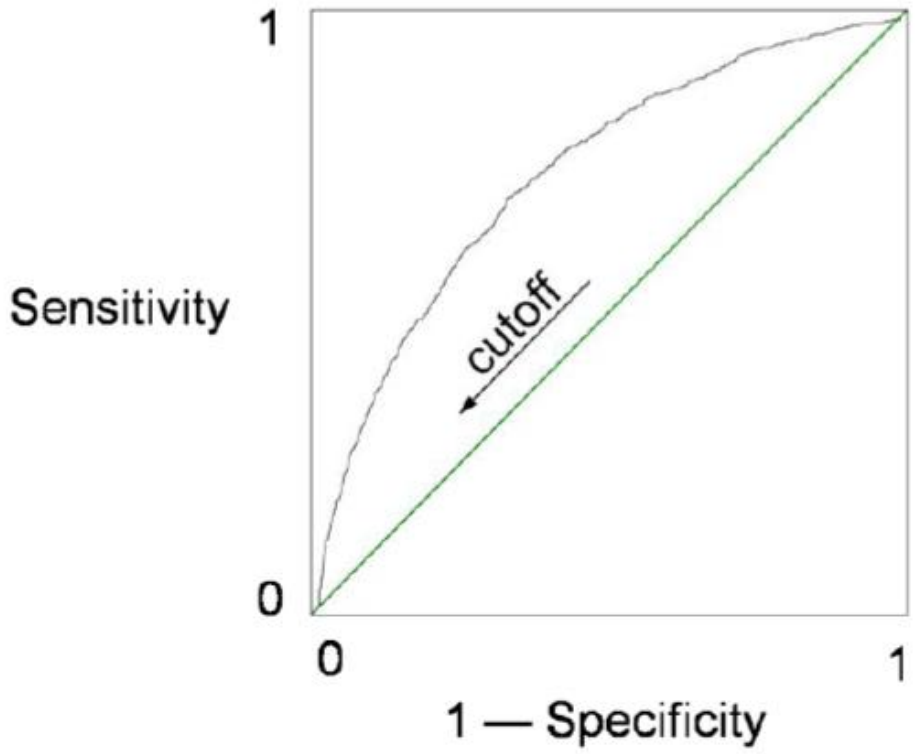
ROC Curve

- The receiver operating characteristic (ROC) curve was adapted from signal detection theory for the assessment of classifiers.
- The ROC curve displays the sensitivity and specificity for the entire range of cutoff values.

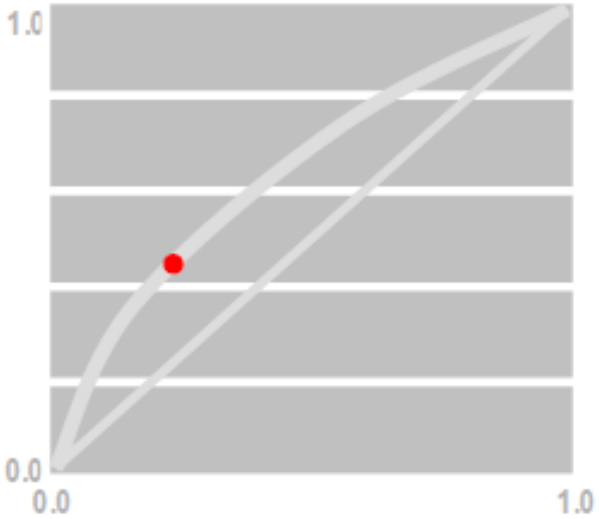


ROC Curve

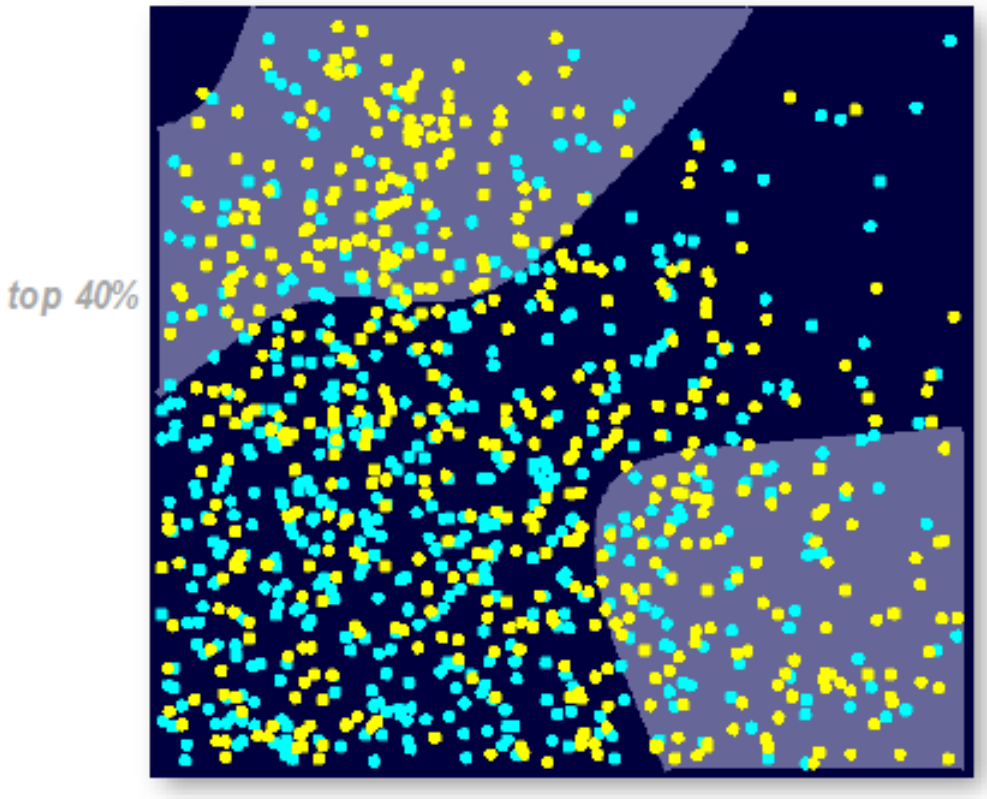
- As the cutoff decreases, more cases are allocated to class 1. Hence, the sensitivity increases and specificity decreases.
- As the cutoff increases, more cases are allocated to class 0. Hence, the sensitivity decreases and specificity increases.
- So, the ROC curve intersects (0, 0) and (1, 1).



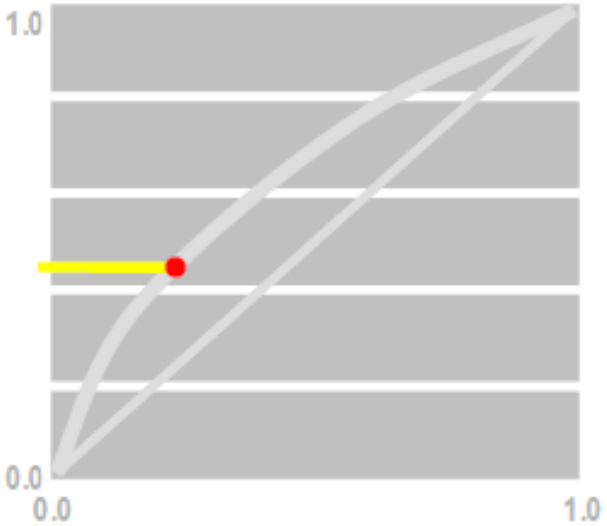
ROC Curve



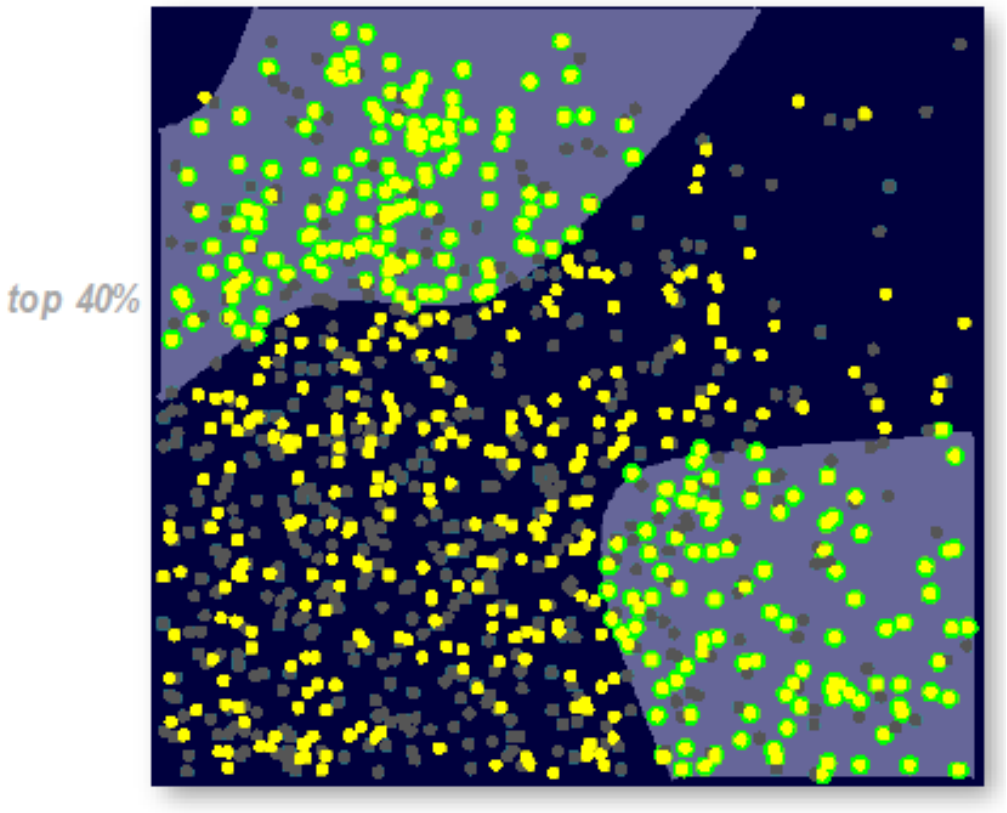
For example, this point on the ROC chart corresponds to the 40% of cases with the highest predicted values.



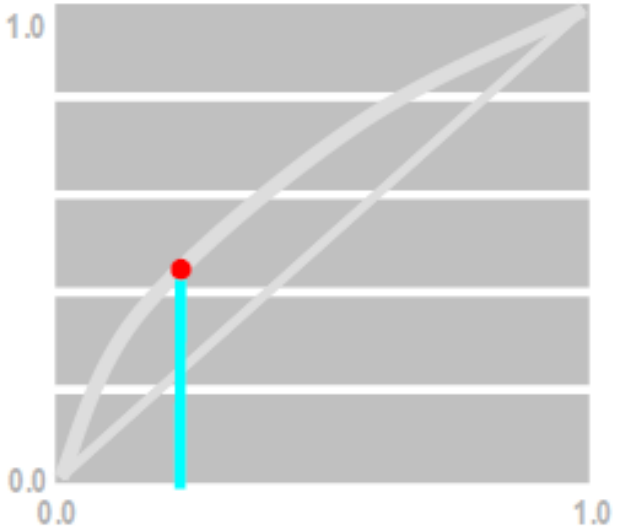
ROC Curve



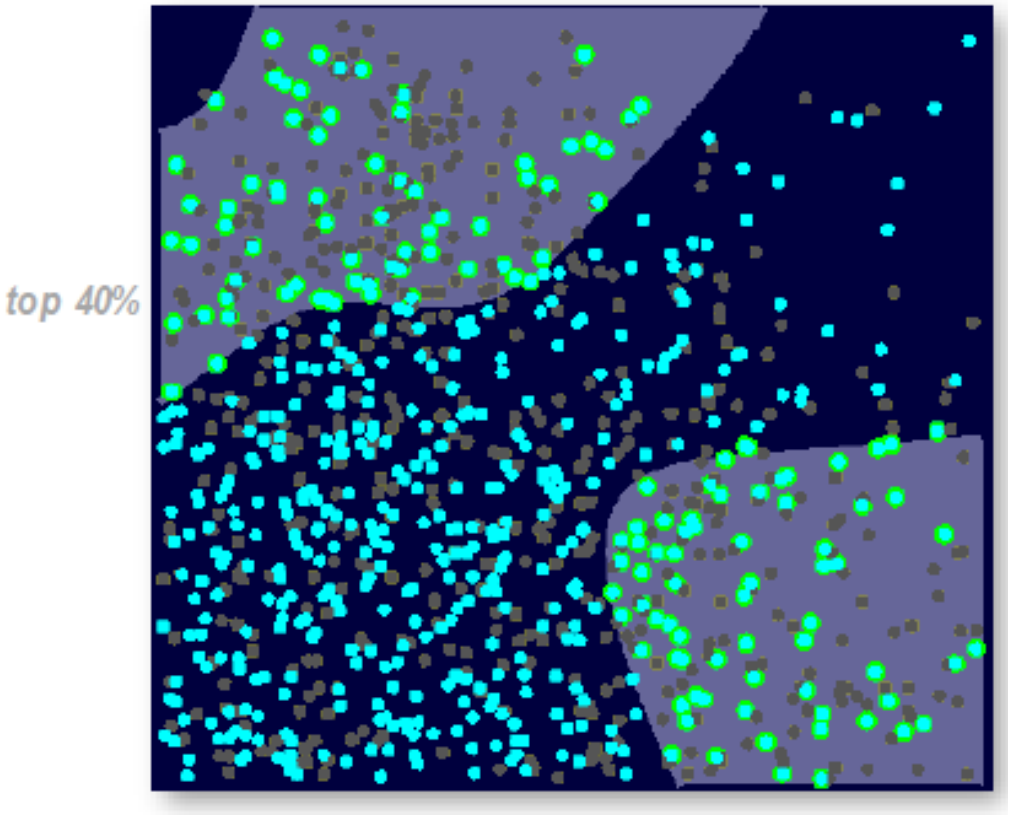
The *y*-coordinate shows the fraction of primary outcome cases captured in the top 40% of all cases.



ROC Curve

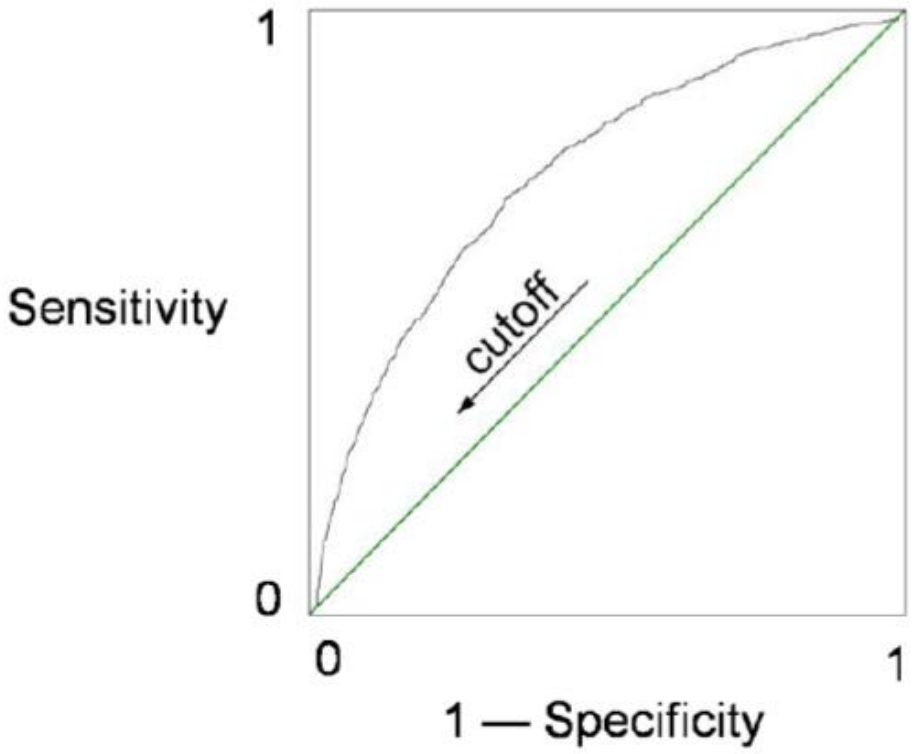


The *x*-coordinate shows the fraction of *secondary* outcome cases captured in the top 40% of all cases.

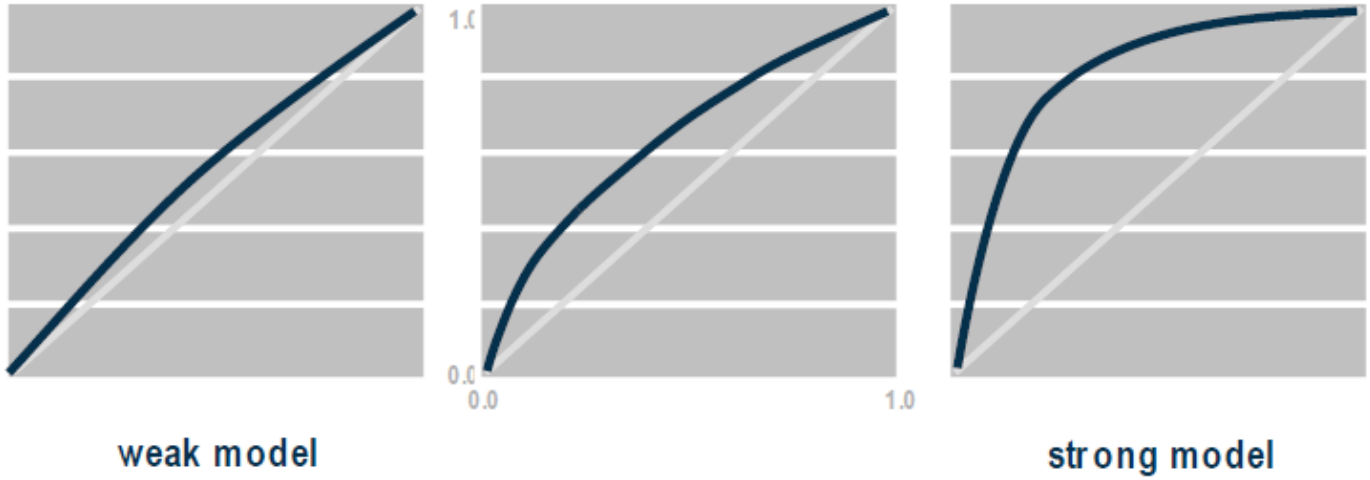


ROC Curve

- If the posterior probabilities were arbitrarily assigned to the cases, then the ratio of false positives to true positives would be the same as the ratio of the total actual negatives to the total actual positives.
- So, the baseline or random model is a 45° angle going through the origin.

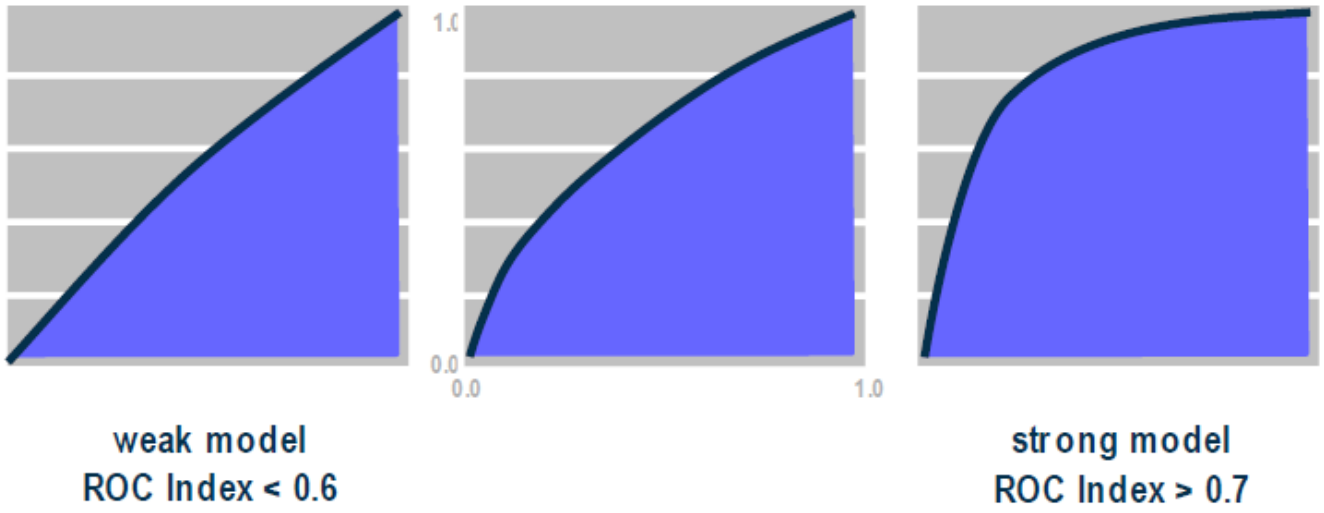


ROC Curve



- As the ROC curve bows above the diagonal, the predictive power increases.
- Models that capture mostly primary outcome cases without capturing secondary outcome cases are strong models.
- A perfect model would reach the (0, 1) point where both sensitivity and specificity equal 1.

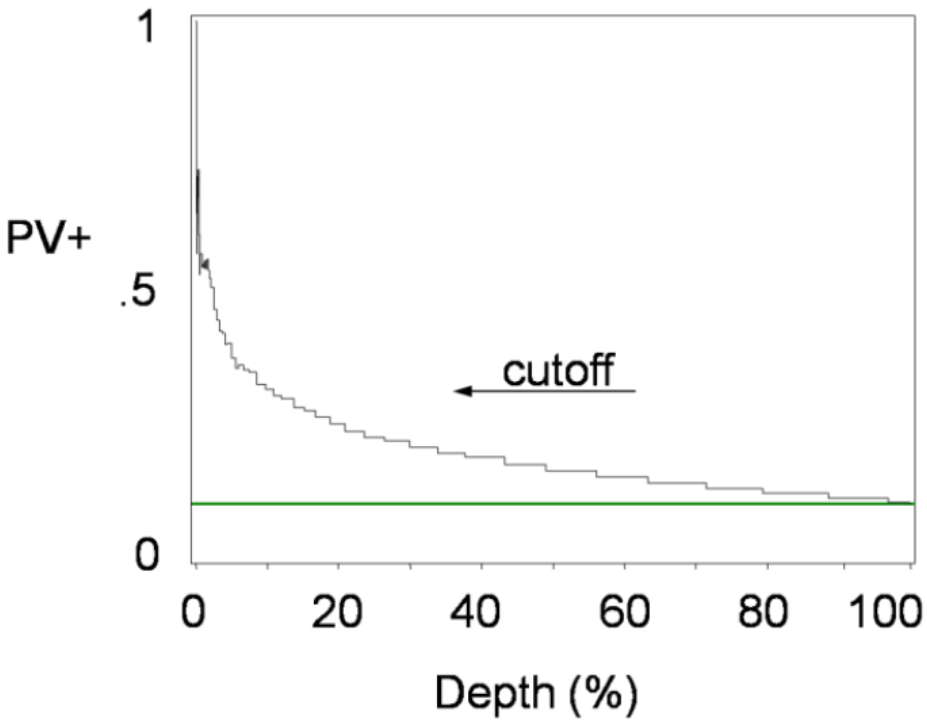
ROC Curve



- The tradeoff between primary and secondary case capture can be summarized by the area under the ROC curve.
- In SAS, this area is called the ROC index. In statistics, it is more commonly called the *c*-statistic.

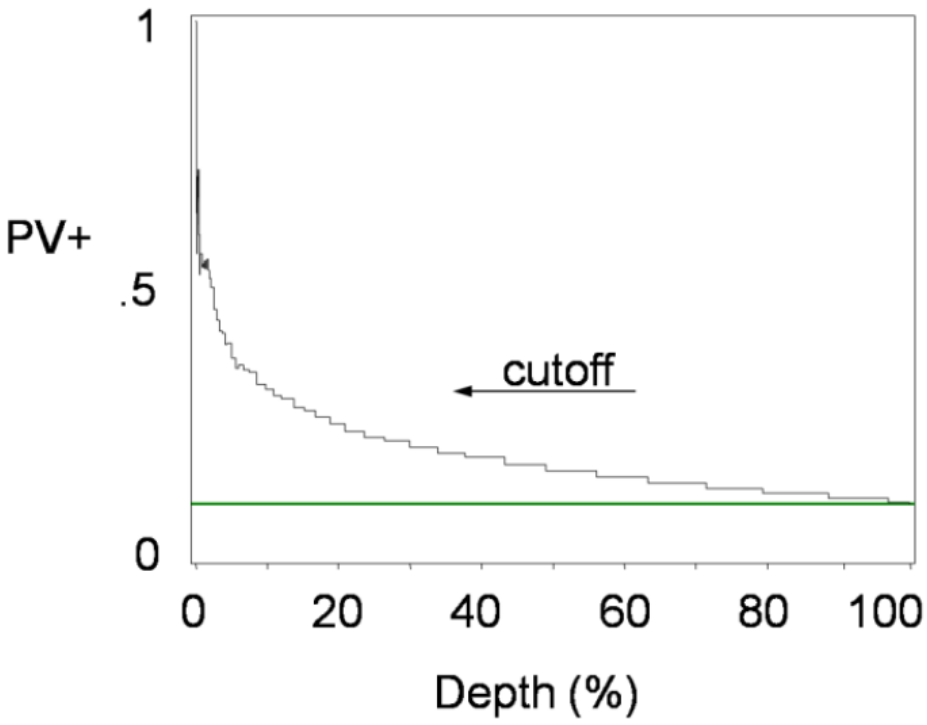
Gains or Response Chart

- The gains chart displays the positive predicted value and depth for a range of cutoff values.
- The depth of a classification rule is the total proportion of cases that were allocated to class 1.
- The simplest way to construct gains chart is to sort and bin the predicted posterior probabilities.



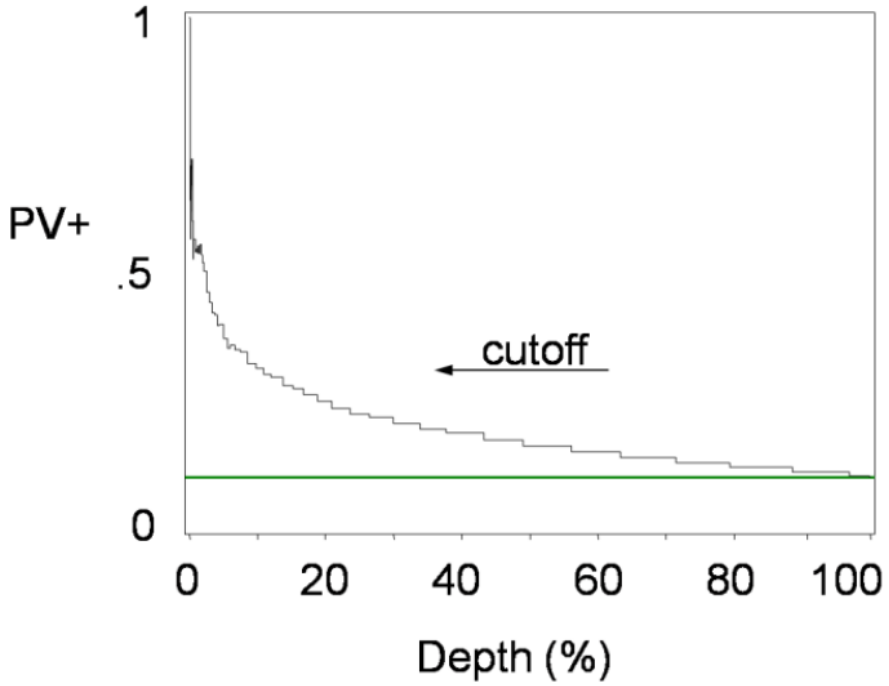
Gains or Response Chart

- As the cutoff decreases, more and more cases are allocated to class 1. Hence, the depth increases and the PV+ approaches the marginal event rate.
- If the posterior probabilities were arbitrarily assigned to the cases, then the gains chart is a horizontal line at the response rate, π_1 (green horizontal line).



Gains or Response Chart

- The gains chart is widely used in database marketing to decide how deep in a database to go with a promotion.
- The lift is $\frac{PV+}{\pi_1}$.
- A plot of sensitivity versus depth is called lift curve, Lorenz curve or concentration curve.
- Lift chart measures how much better a predictive model comparing with a naive model.



Lesson Summary

- A confusion matrix is used to assess the model with a binary target.
- The outcomes of a confusion matrix can be categorised as True positive, True negative, False positive or False negative. The event of interest is denoted as positive.
- The performance statistics derived from a confusion matrix are accuracy, misclassification rate, sensitivity, specificity, positive predicted value and negative predicted value.
- The statistical charts produced from these performance statistics are the ROC curve, gains or response chart and lift chart.

How do we compare the models?



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Topic 10.3
Model Assessment and Models Comparison

Models Comparison

By Dr Lee How Chinh



Objectives

- Define the model fit statistics based on the prediction type for model comparison.

Models Comparison

Model Comparison Node

- The Model Comparison node is a Model Studio node that is automatically added to a pipeline when a Supervised Learning node is also added.
- The Model Comparison node belongs to the Assess category in the SAS Big Data approach of Explore, Modify, Segment, Model and Assess (**EMSMA**).
- The Model Comparison node can only compare the performance of Model nodes in the same pipeline.

Model Comparison Node

- The Model Comparison node enables you to compare the performance of models using various performance statistics.
- The pipelines comparison tab compares the champion models among the pipelines.
- The performance statistics can be grouped by prediction type, decision, ranking, and estimate predictions.

Model Comparison Node

For decision prediction,

- Accuracy or misclassification
  - It tallies the correct or incorrect prediction decisions.
- Cumulative lift
  - It compares the performance of the model to a random selection.
- The Kolmogorov-Smirnov (KS) statistic
  - This statistic describes the ability of the model to separate the primary and secondary outcomes.

Model Comparison Node

For ranking prediction,

- ROC index
  - This index equals the percent of concordant cases plus one-half times the percent tied cases.
- Gini coefficient
  - This coefficient is a metric that indicates the model's discriminatory power.
  - This coefficient equals 2(ROC index – 0.5).

Model Comparison Node

For estimate prediction,

- Average squared error
  - ASE is the average squared difference between the estimated and actual values.
  - It measures, on average, how close the predicted estimates and actual values are.

- You can refer to this reference link for the details of the Model Comparison Node.

Lesson Summary

- A model comparison node from Assess compares the performance of supervised models in a pipeline.
- The performance statistics are used based on prediction type.
- Accuracy or misclassification, cumulative lift, and the KS statistic are for decision prediction.
- The ROC index and Gini coefficient for ranking prediction.
- Average square error for estimates prediction.

Let us explore all these on SAS Viya.



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Topic 10.4
Model Deployment

Scored data requirement

By Dr Lee How Chinh



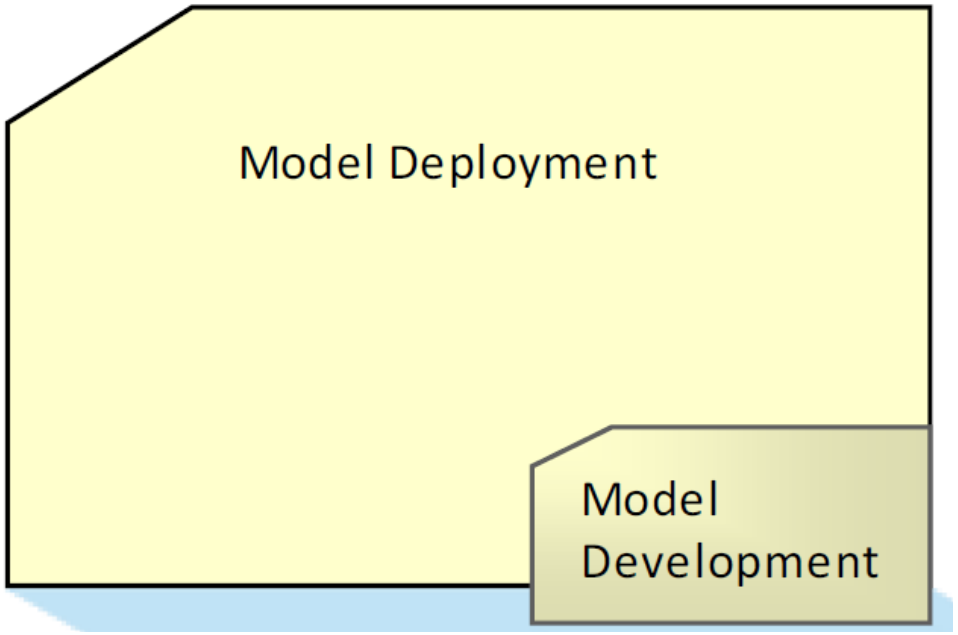
Objectives

- Explain the concepts of scoring
- List the requirements for scoring new data.

Scoring Predictive Models

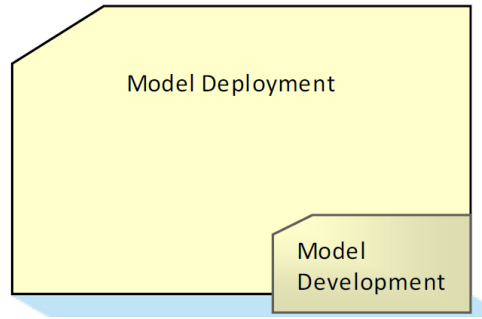
Scoring Predictive Models

- The predictive modelling task is incomplete if only a model and allocation rule are determined.
- The model must be practically applied to new cases.
- This process is called scoring.



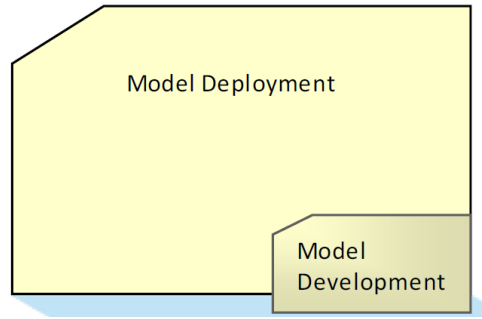
Scoring Predictive Models

- The scoring process can be tremendously burdensome in database marketing because the data to be scored might be many times more massive than the data used to develop the model.
- Moreover, the data might be stored in a different format on another system using other software.
- The model might need to be integrated into an online monitoring system in other applications, such as fraud detection.



Scoring Requirements

- The model results in a formula or rules.
- A comparable scoring data.
  - The scoring data that modified according to the training data set.
- The scoring code
  - You do not rerun the algorithm; apply score code (equations) obtained from the final model to the scoring data.



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Topic 10.5
Model Deployment

Model implementation

By Dr Lee How Chinh



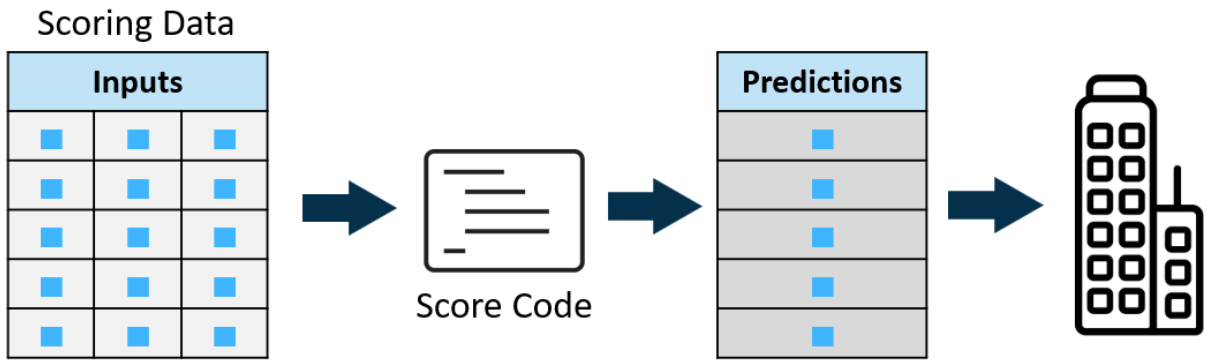
Objectives

- Explain model implementation
- Use the SAS Viya to score the models.

Model Implementation in SAS Viya

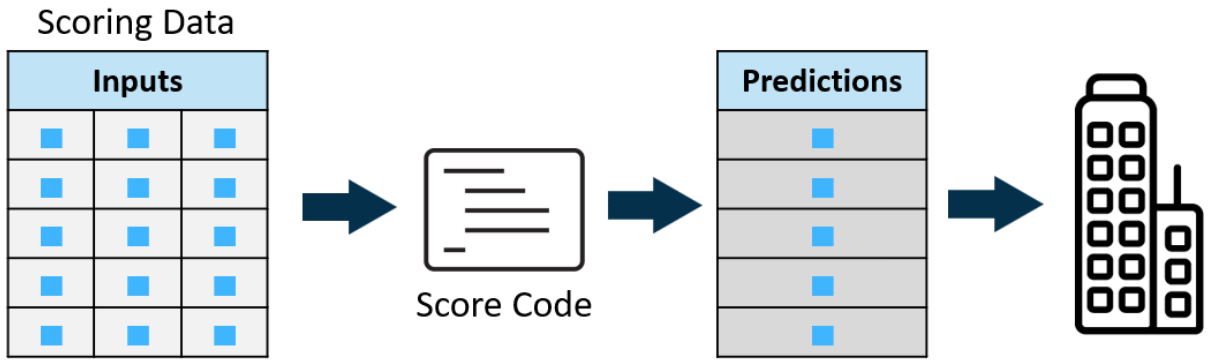
Model Implementation

- The scoring process is sometimes referred to as model deployment, model production, or model implementation.
- For scoring, the model is first translated into another format (typically, score code).

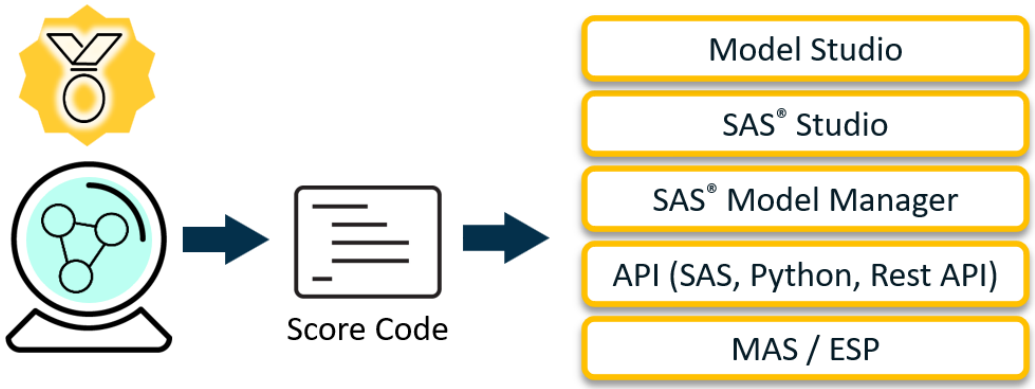


Model Implementation

- In the SAS Viya environment, score code is an SAS program that can easily be run on a scoring data set.
- The model is then applied to the scoring data set to obtain predicted outcomes.
- Based on the predictions from the model, the enterprise makes business decisions and takes action.



Model Implementation



- Model Studio creates two types of SAS language score code for the purpose of scoring new data:
  - DATA Step
  - Analytic Store (ASTORE)

- To generate score code for an entire pipeline, the score code for each node producing Node score code is appended together into a single DATA step.
- When the nodes in a pipeline produce multiple analytic stores, or one or more analytic stores and DATA step score code, an EP score code file is created. EP score code represents the score code produced by these pipelines

Model Management Using SAS Model Manager

- SAS Model Manager streamlines analytical model management and administration.
- SAS Model Manager provides a web-based environment to support life cycle management and governance of models.
- SAS Model Manager:
  - Store SAS and open-source models within projects or as stand-alone models.
  - Develop and validate candidate models.
  - Assess and compare candidate models for champion model selection.
  - Then publish and monitor champion and challenger models to ensure optimal model performance.

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Topic 10.6
Model Deployment

Monitoring and Updating the Model

By Dr Lee How Chinh

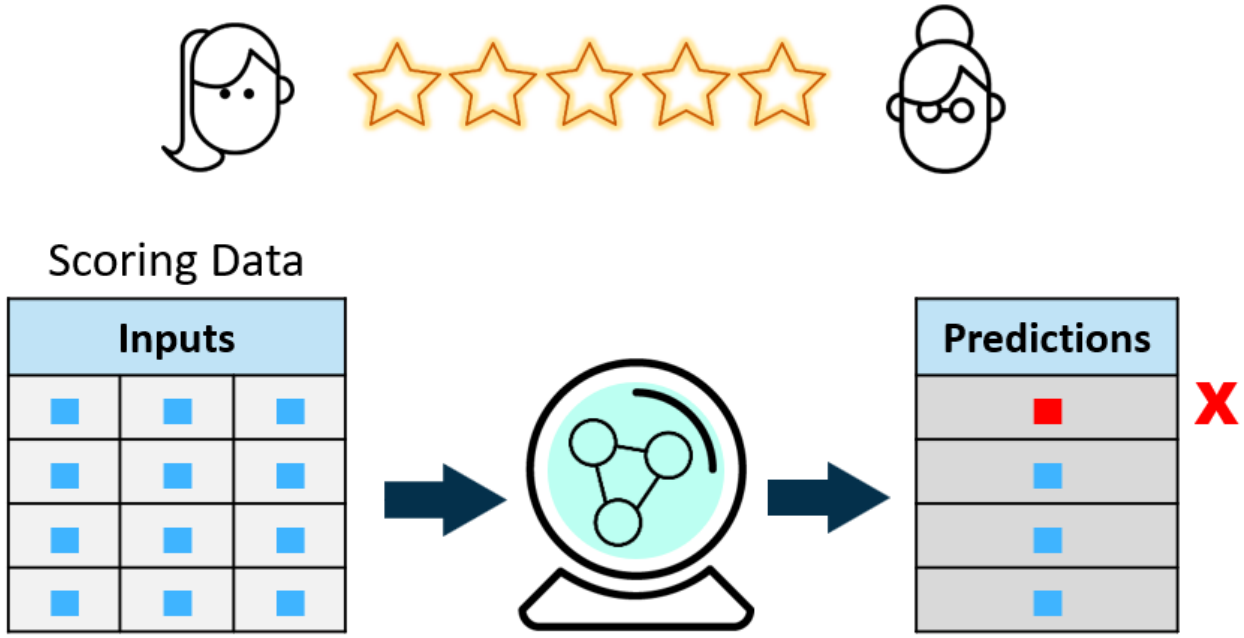


Objectives

- Describe model monitoring and updating.
- Apply model updating.

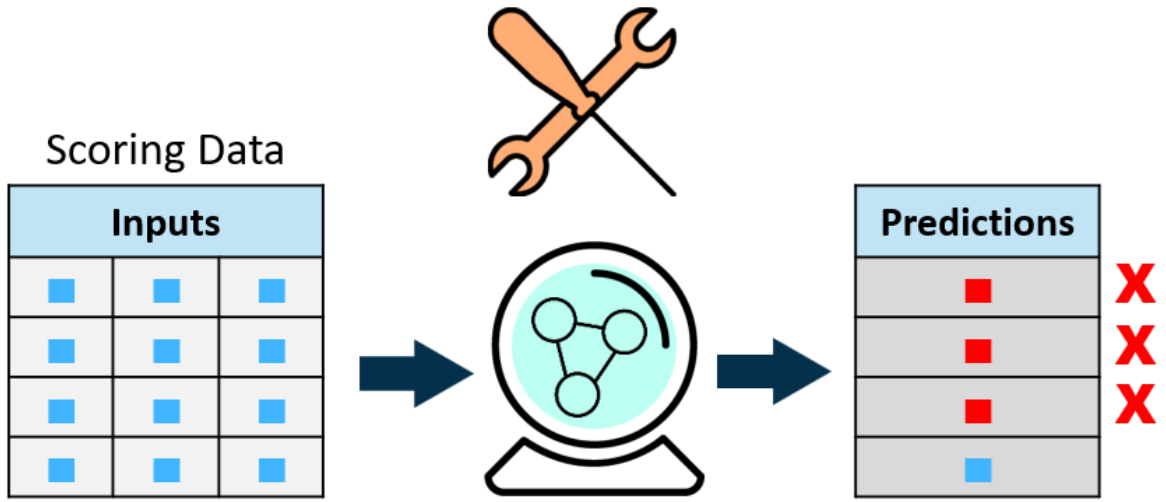
Monitoring and Updating the Model

- After a model has been deployed, it must be monitored by closely monitoring its accuracy and performance over time.
- A model must be revisited and re-evaluated in light of new data and changing circumstances



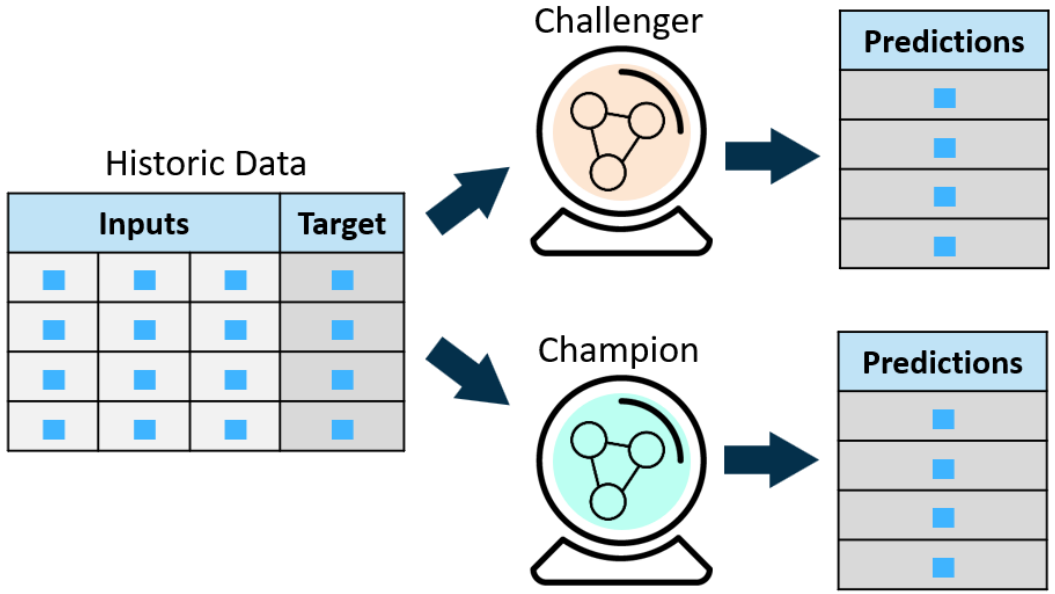
Monitoring and Updating the Model

- Models are trained on static snapshots of data.
- After the model is deployed, the environment inevitably changes and becomes less and less like the conditions that were captured in the training data.
- As a result, the model's predictions typically become less accurate over time.



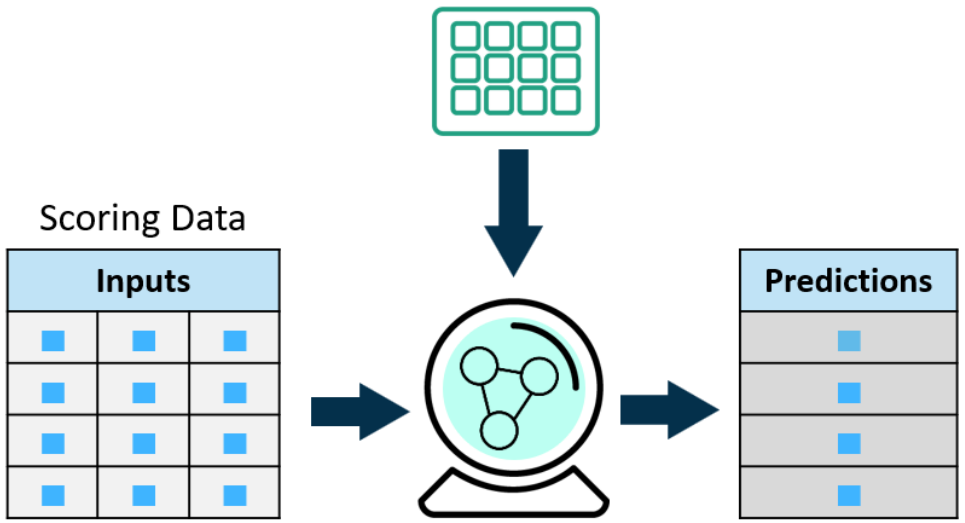
Champion-Challenger Testing

- Champion-challenger testing is a common model deployment practice.
- This method compares the performance of a new challenger model with the performance of the deployed model on a historic data set at regular time intervals.
- If the challenger model outperforms the deployed model, the challenger model replaces it. The champion-challenger process is then repeated.



Online Updates

- Another approach to refreshing a trained model is through online updates.
- Online updates continuously change the value of model parameters or rules based on the values of new, streaming data.
- It is prudent to assess the trustworthiness of real-time data streams before implementing an online modelling system.



How to use SAS Viya for scoring and model monitoring?



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